Property Taxes and Residential Mobility: Evidence from São Paulo

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Abstract

Property taxes are a key policy tool that shapes housing markets, affecting affordability, homeownership, and residential sorting. When property taxes rise, do existing homeowners respond by moving, and if so, who moves? We study this question using São Paulo's 2014 property tax reform, which generated substantial variation in tax changes among properties on the same city block. Within-block comparisons allow us to estimate the effect of tax increases on mobility without the endogeneity that arises when tax variation reflects differences in the provision of local public goods. We find that property tax increases induce residential mobility. Furthermore, linking property records to matched employer-employee data, we find heterogeneity in the mobility response by owner characteristics, with the effect concentrated among older owners. Our results provide the first causal evidence on who is responding to property tax increases and the first evidence on tax-induced mobility in a developing country.

JEL: H71, R21, H31

Keywords: property tax, residential mobility, homeownership, housing, tax reform

1 Introduction

Policymakers often consider property taxes as a tool to pursue a range of policy objectives. Since property taxes affect the cost of homeownership, and thus housing prices, they have been proposed as a means to improve housing affordability.[1] At the same time, because property taxes are a major source of municipal revenue, increases have also been proposed to expand funding for local public goods without introducing regressivity.[2] Yet despite how frequently property tax reforms are debated, a basic empirical question remains open: when property taxes rise, do existing homeowners respond by selling, and if so, who among them is most likely to move? This question matters because a household's decision to own a home in a specific location has extended consequences. For most households, housing is their primary asset, and a tax-induced decision to stay or sell is a decision about their largest financial holding (Campbell, 2006; Artigue et al., 2025). A household's location also determines its access to local public goods, schools, and labor markets (Bayer et al., 2007; Chetty and Hendren, 2018). If tax increases induce some owners to stay but not others, they reshape the composition of neighborhoods and the dynamics of cities (Guerrieri et al., 2013).

A growing literature studies how property taxes affect residential mobility. In the United States, the most studied reform is California's Proposition 13, which reduces property taxes for existing owners by capping assessed values at purchase-year levels. Ferreira (2010) and Wasi and White (2005) exploit this setting and show that lower property taxes reduce mobility. Similarly, Fraenkel (2024) uses Ohio's mass reappraisals and finds that property tax increases induce mobility. Yet even if taxes affect mobility on average, a harder question is who responds. Coven et al. (2025) develop an overlapping generations model in which higher property taxes disproportionately induce elderly homeowners to sell. Empirically, several papers find evidence of heterogeneous mobility responses by age (Shan, 2010; Giesecke and Mateen, 2022; Nielsson et al., 2024), but identification in each case relies on between-municipality variation in tax rates. Because property tax revenue funds local public goods, municipalities that raise taxes may simultaneously improve provision of public goods. Thus, relying on between-municipality variation alone makes it difficult to separate the causal effect of the property tax on household behavior, as municipalities with different tax rates may also differ in school quality, public safety, and other amenities that affect residential decisions.

We provide this evidence by exploiting São Paulo's 2014 property tax reform, which

[1] In South Korea, policymakers have considered increasing property taxes to cool soaring housing prices and improve affordability in Seoul.

[2] In New York City, the city government has debated whether raising property taxes can increase revenue while limiting regressivity.

simultaneously updated tax assessment parameters, generating substantial variation in tax changes even among properties on the same city block. São Paulo's large and dense housing market enables block-level fixed effects that absorb local public goods and amenities, allowing us to estimate the effect of tax increases on mobility without the endogeneity that arises when tax variation reflects differences in public goods provision. A distinctive feature of our setting is the ability to link property records to RAIS, Brazil's matched employer-employee dataset, which allows us to observe owner age and income. We estimate the effect of tax increases on mobility using OLS with block fixed effects and flexible property controls, and validate the identifying assumption with a pre-reform placebo. Our sample comprises approximately 296,000 apartments, with a RAIS-matched subsample of approximately 54,000.

We find that property tax increases induce residential mobility. A 10 percentage point tax increase raises the six-year transaction probability by 0.86 percentage points, or 4.5 percent relative to the 14.4 percent baseline rate. Pre-reform mobility is unrelated to the tax increase, supporting the identifying assumption. Linking property records to matched employer-employee data, we show that the mobility response is concentrated among older owners: the differential effect increases monotonically with age and is stable across all control specifications. This pattern is consistent with the theoretical and empirical literature on mobility among existing homeowners.

This paper contributes to the literature in three ways. First, we provide the first causal estimates of who responds to property tax increases. Our within-block identification separates the effect of tax increases from the provision of local public goods, and our linkage to matched employer-employee data allows us to observe the demographics and employment status of the owners who sell. Second, we document a heterogeneous mobility response by age, providing empirical evidence that is consistent with existing theoretical models. Third, we provide the first evidence on tax-induced mobility in a developing country, using administrative data from the largest city in South America.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and the 2014 reform. Section 3 presents the data. Section 4 details the empirical strategy. Section 5 presents the main results and heterogeneity analysis. Section 6 discusses distributional consequences, limitations, and concludes.

2 Institutional Background

2.1 Residential Property Taxes in São Paulo

Brazilian municipalities levy the *Imposto Predial e Territorial Urbano* (IPTU), a property tax on all urban properties. In the city of São Paulo, IPTU is a major source of municipal revenue, accounting for approximately 15 percent of the city's total revenue.[3] For individual households, the tax represents a nontrivial financial obligation: the mean annual tax liability in our sample is BRL 2,725, equivalent to approximately 66 percent of the average monthly wage in the state of São Paulo.[4]

IPTU is levied on the assessed value, not the market value. The city government sets and updates assessed values annually using a formula based on observable property characteristics and tax parameters. The assessed value of each property is the sum of a construction component and a land component (Appendix A displays the tax form):

$$V_i = \underbrace{C_i \times \phi^C_{z(i)} \times \delta_i}_{\text{construction}} + \underbrace{L_i \times \phi^L_{b(i)} \times f_i}_{\text{land}} \tag{1}$$

where C_i is the construction area, $\phi^C_{z(i)}$ is the construction price per square meter in property i's fiscal zone z (uniform within zone for a given construction type), δ_i is an obsolescence factor based on building age, L_i is the lot area (scaled by the condominium's ideal fraction for apartments), $\phi^L_{b(i)}$ is the land price per square meter assigned to property i's block b (varying at the street frontage level), and f_i collects additional adjustment factors for lot depth, terrain type, and condominium structure.

Tax liability is then computed by applying a progressive rate schedule to the assessed value:

$$\tau_i = T(V_i) \cdot V_i \tag{2}$$

where $T(\cdot)$ is the effective tax rate, determined by the bracket into which V_i falls. The progressive structure means that properties with higher assessed values face higher marginal rates. Appendix A reports the complete pre- and post-reform rate schedules.

Two features of this system are important for our purposes. First, the IPTU formula is a rigid, government-set function of observable property attributes. The resulting tax liability for each property is deterministic: owners cannot manipulate their assessments, and the municipality does not exercise discretion in applying the formula. Second, because the

[3] Total IPTU revenue for the city of São Paulo was BRL 16.4 billion (approximately USD 2.7 billion) in 2024 (PMSP, 2024).

[4] BRL 4,141 in 2014 (IBGE PNAD Contínua; https://sidra.ibge.gov.br/tabela/5439). The national minimum wage was BRL 724 per month in 2014.

formula is nonlinear and depends on multiple property characteristics (area, age, construction standard), even two apartments on the same block, sharing the same land price ($\phi^L_{b(i)}$) and construction price ($\phi^C_{z(i)}$), can experience different tax increases when assessment parameters are updated based on the property characteristics. These formula-driven differences generate within-block variation in tax changes that is central to our identification strategy.

2.2 The 2014 Property Tax Reform

In 2013, the São Paulo City Council approved a comprehensive reform of the IPTU assessment system under the Fernando Haddad administration. The reform was implemented beginning in 2015. To limit the immediate burden on taxpayers, the municipality capped annual increases in tax liability at 10 percent for residential properties, phasing in larger adjustments over several years.[5] Its stated objective was to update property assessments, which had not been systematically revised in over a decade, while introducing a progressive structure to limit the burden on lower-income households in peripheral areas.

The reform revised essentially all elements entering the tax formula simultaneously:

1. **Land price assessments** ($\phi^L_{b(i)}$): Updated to reflect current market conditions. Central neighborhoods received substantially higher valuations, while some peripheral areas saw more moderate adjustments, as Figure 1 shows.

2. **Construction price assessments** ($\phi^C \to \phi^C_{z(i)}$): Before the reform, ϕ^C was uniform city-wide for each building type. The reform introduced a spatial discontinuity in fiscal zones, dividing the city into three fiscal zones (ZF-1, ZF-2, and ZF-3, from central to peripheral) to reflect construction assessments closer to actual market values. ZF-1 encompasses the wealthiest, most central districts and receives the highest $\phi^C_{z(i)}$. After the reform, construction prices vary substantially across the city, as Figure 2 illustrates.

3. **Tax rate schedule**: New brackets and rates were introduced. The post-reform residential rate schedule ranges from 0.7 to 1.5 percent of assessed value, compared to a narrower range under the pre-reform schedule (Appendix A).

Because the reform updated all assessment inputs simultaneously, the resulting tax change for each property depended on the complex interaction of its pre-reform characteristics (built area, land area, construction standard, age, location) with the new parameters. Properties with identical pre-reform taxes could experience very different increases depending on the composition of their assessed value between land and construction, their position

[5] Lei 15.889/2013, Art. 9.

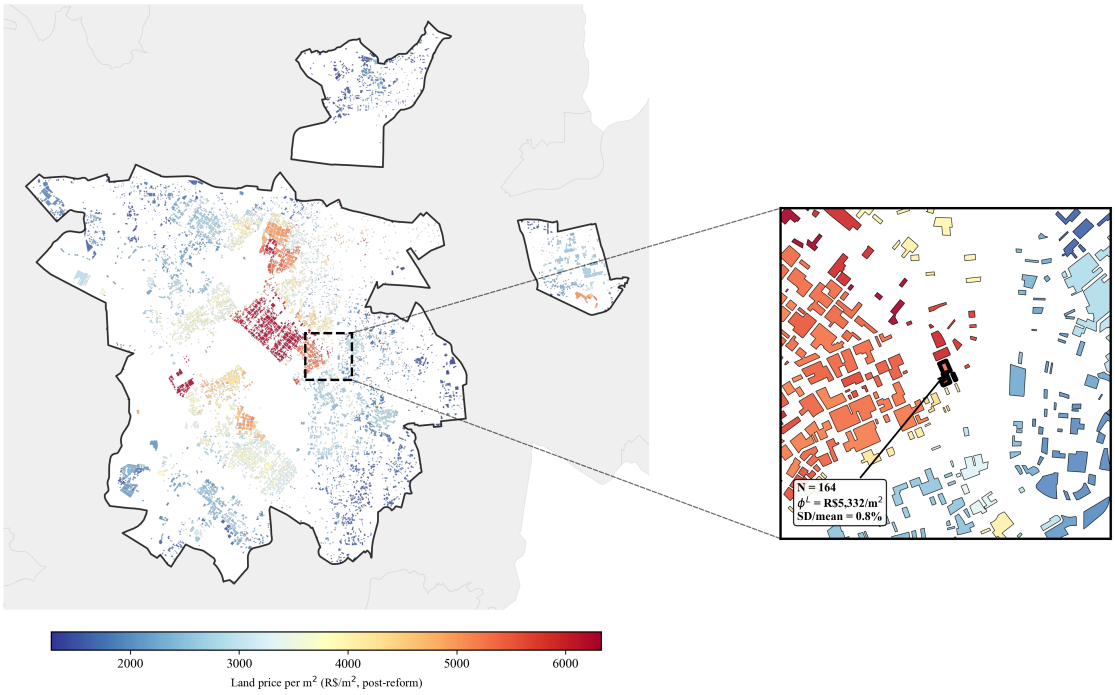


Figure 1: Land prices vary by block

Notes: Post-reform land prices per m^2 for apartments in ZF-1. Land prices vary continuously at the block level but are very similar within blocks: 44 percent of blocks have zero within-block variation, and the median SD/mean is 0.5 percent (Figure B.1 in Appendix B.1). The inset zooms into a single block (bold black lines), which contains $N = 164$ apartment units with a land price of $\phi^L_{b(i)}$ = R\$5,332/m^2 and a within-block SD/mean of $\phi^L_{b(i)}$ of only 0.8 percent.

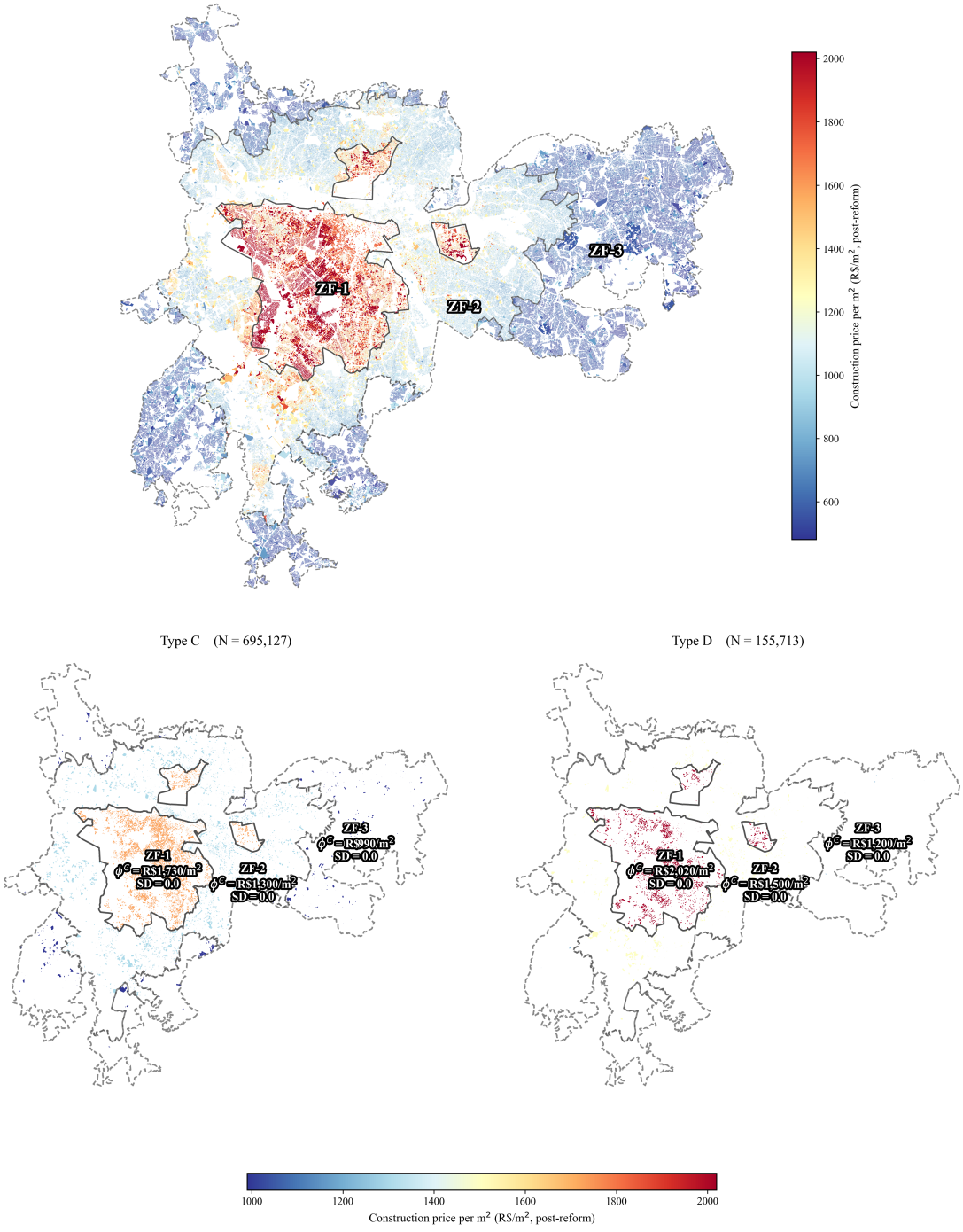


Figure 2: Construction prices vary by fiscal zone and building type

Notes: Upper panel shows post-reform construction prices per m^2 across the entire city of São Paulo, pooling all building types. Construction prices are uniform within each fiscal zone (ZF-1, ZF-2, ZF-3) for a given construction type but shift discretely at zone boundaries. Lower panel shows construction prices separately by building type (Type C and Type D). Within a given fiscal zone and building type, construction prices are fixed (SD = 0).

in the new rate brackets, and the magnitude of the land price update in their specific location. Figure 3 shows the resulting distribution of tax increases for ZF-1 apartments: the reform raised tax liabilities by 53 percent on average, with substantial heterogeneity ranging from near-zero to over 100 percent. This within-block heterogeneity forms the basis of our empirical strategy. We examine the distributional consequences of the reform in Section 6.

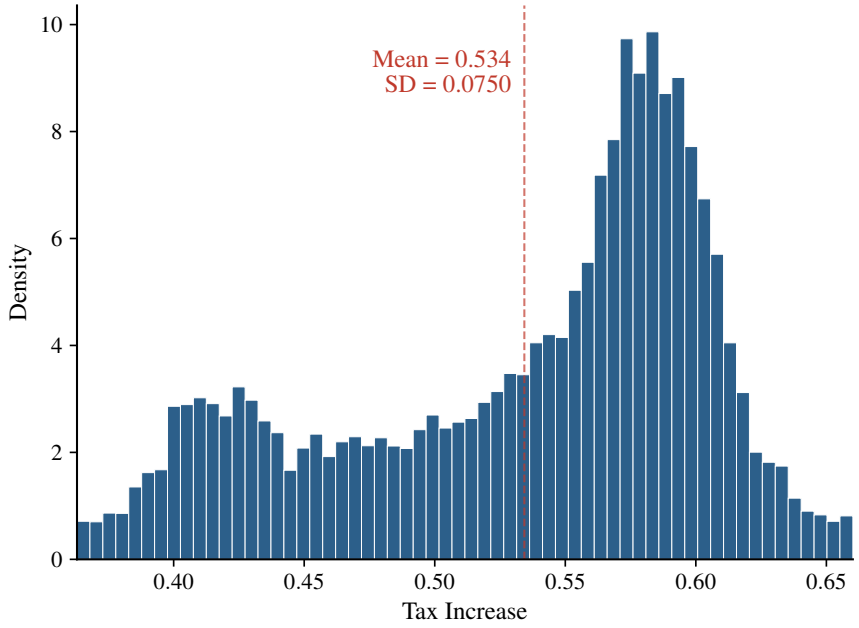


Figure 3: Tax increases vary substantially across properties

Notes: Distribution of percentage changes in tax liability from 2014 to 2015 for ZF-1 apartments. Sample trimmed at the 1st and 99th percentiles. $N \approx 296{,}000$.

3 Data

3.1 Data Sources

We combine three administrative datasets. The first two are maintained by the São Paulo Municipal Secretariat of Finance (*Secretaria Municipal da Fazenda*) and together provide near-universal coverage of the city's housing stock and transaction activity. The third is a federal matched employer-employee dataset that provides information on property owners' demographics and earnings.

Property tax registry (IPTU). The IPTU dataset records the annual tax liabilities and assessed characteristics of all properties in São Paulo. We use records from 2014, 2015, and 2020, corresponding to the pre-reform, post-reform, and long-term post-reform years.

For each property, the registry contains geographic coordinates, land area, construction area, construction standard, building age, and fiscal zone classification. A unique property identifier allows consistent tracking across years. Because the assessment formula is a deterministic function of observable property attributes and government-set parameters, we can back-engineer each property's tax liability under both the pre-reform and post-reform rules. This allows us to compute the reform-induced tax change for every property in the sample, yielding the treatment variable described in Section 4.

Transfer tax registry (ITBI). The ITBI dataset records all property transfer tax filings (*Imposto sobre Transmissão de Bens Imóveis*) in São Paulo, covering transactions from 2006 to 2024. For each transaction, it reports the declared sale price, transaction date, and property identifiers. We merge ITBI records to IPTU using the unique property identifier to determine whether each property was transacted during the post-reform period. We restrict to market transactions (*compra e venda*), excluding donations, inheritance transfers, and judicial awards.

Employer-employee records (RAIS). To examine heterogeneity by owner characteristics, we link property owners to Brazil's *Relação Anual de Informações Sociais* (RAIS), a matched employer-employee dataset covering the universe of formal sector workers. We use the 2014 RAIS wave, the last year before the reform took effect, so that owner characteristics are measured prior to the tax change. RAIS provides information on wages, age, education, and employment history. We match IPTU property owners to RAIS records using both partially masked CPF (the Brazilian individual taxpayer identification number, partially masked for privacy) and owner name. The match rate is approximately 18 percent (N = 54,512). Because RAIS covers only formal sector workers, the matched subsample excludes informal workers, retirees, and the self-employed.

3.2 Sample Construction

We make two sample restrictions. First, we focus on ZF-1, which covers the wealthiest, most central districts of São Paulo. Property values in ZF-1 are well above the exemption thresholds that apply to lower-valued properties elsewhere in the city, and ZF-1 experienced the largest tax increases from the reform, providing substantial variation in the treatment variable. Second, within ZF-1, we restrict to apartments rather than single-family houses. Apartments constitute a relatively homogeneous property type, reducing concerns about unobserved heterogeneity across fundamentally different housing structures, and account for the vast majority of residential properties in central São Paulo. The full sample comprises 296,076 apartments.

The outcome variable, mobility, is a binary indicator equal to one if the property was transacted (as recorded in the ITBI) during the post-reform period 2015–2020. We also construct a placebo outcome equal to one if the property has any ITBI transaction during 2009–2013, the pre-reform period.

The treatment variable is the percentage change in tax liability from 2014 to 2015: $(\tau_{2015} - \tau_{2014})/\tau_{2014}$. After the initial reform in 2014, tax increases accumulated over subsequent years as the land and construction price parameters (ϕ^L and ϕ^C) were updated to match inflation. This first-year change therefore captures only a fraction of the reform's full adjustment for most properties. As a robustness check, we also examine the cumulative tax increase from 2014 to 2020.

For the heterogeneity analysis, we use the RAIS-matched subsample of 54,512 apartments whose owners were successfully linked to formal employment records. We restrict this subsample to owners aged 18–100 with positive wages.

3.3 Descriptive Statistics

Table 1 presents summary statistics. The average ZF-1 apartment has a construction area of 182 m^2 and a pre-reform assessed value of BRL 249,117, with annual pre-reform tax liability averaging BRL 2,726 (approximately 1.1 percent of assessed value). The reform raised mean tax liability to BRL 4,093, an average increase of 53 percent. Figure 3 shows the full distribution, which exhibits substantial heterogeneity across properties.

The post-reform mobility rate is 14.4 percent: roughly one in seven ZF-1 apartments changed hands between 2015 and 2020.

Property characteristics in the RAIS-matched subsample are comparable to the full sample in terms of tax increase and post-reform mobility, though RAIS-matched properties tend to be somewhat smaller and newer, as Table 1 reports. The subsample differences are expected given that RAIS captures only formal sector employees. Crucially, the treatment variable and the outcome are nearly identical across the two samples, and the spatial distribution is similar, as shown in Appendix B.2. Owners in this subsample earn BRL 14,636 per month on average (median BRL 11,276). The mean owner age is 50 years, and 90 percent hold a college degree or higher.

Table 1: Summary Statistics

	Full Sample		RAIS-Matched	
	Mean	(SD)	Mean	(SD)
Panel A: Property Characteristics				
Construction area (m^2)	181.9	(115.1)	169.0	(91.3)
Land area per unit (m^2)	32.5	(24.2)	29.7	(19.4)
Property age (years)	34.9	(14.3)	30.9	(14.0)
Assessed value, pre-reform (BRL)	249,117	(253,087)	220,665	(189,276)
Tax liability, pre-reform (BRL)	2,726	(3,686)	2,314	(2,677)
Tax liability, post-reform (BRL)	4,093	(5,393)	3,474	(3,896)
Tax increase (%)	53.4	(7.5)	53.2	(7.4)
Ownership transferred, 2015–2020 (%)	14.4	(35.1)	13.8	(34.5)
Panel B: Owner Characteristics				
Monthly wage (BRL)	–	–	14,636	(12,589)
Owner age (years)	–	–	50.1	(10.9)
Male (%)	–	–	62.8	(48.3)
College degree or higher (%)	–	–	90.2	(29.7)
Observations	296,076		54,512	

Notes: Full sample includes all apartments in Fiscal Zone 1 (ZF-1). RAIS-matched sample includes properties whose owners were linked to formal-sector employment records, restricted to owners aged 18–100 with positive wages. Tax increase is the percentage change in tax liability from the 2014 reform. Ownership transfer is measured using ITBI (property transfer tax) transaction records.

4 Empirical Strategy

4.1 Specification

We estimate the effect of the reform-induced tax increase on residential mobility using OLS:

$$Y_i = \beta \cdot \text{TaxIncrease}_i + f(X_i) + \alpha_j + \varepsilon_i, \tag{3}$$

where Y_i is a binary indicator for whether property i was transacted between 2015 and 2020, TaxIncrease_i is the percentage change in tax liability from 2014 to 2015, $f(X_i)$ is a flexible function of property controls, and α_j are block-level fixed effects. The coefficient β measures the marginal effect of a larger tax increase on the probability of mobility.

The fixed effects operate at the block level, so the identifying variation comes from comparisons among properties on the same block. Within a block, properties are exposed to essentially the same local public goods, school quality, and neighborhood amenities, making it unlikely that these factors confound the estimated relationship. Standard errors are heteroskedasticity-robust and clustered at the neighborhood level. The controls X_i include the observable property characteristics that enter the tax assessment formula and may independently predict mobility: construction area, land area, property age, and construction standard. In the main specification, all continuous controls are binned at deciles, providing a flexible nonparametric fit. We report robustness to three alternative specifications: linear controls, quintile bins, and B-spline bases.

As a falsification test, we estimate the identical specification replacing the outcome with pre-reform mobility, a binary indicator equal to one if property i was transacted between 2009 and 2013. If the tax increase captures unobserved property characteristics that persistently predict turnover, it should predict mobility in both periods. A null placebo coefficient would indicate that the post-reform relationship is not driven by such confounders.

4.2 Identification

Credible estimation of β requires that, conditional on block fixed effects and property controls, residual variation in tax increases is uncorrelated with unobserved determinants of mobility. The structure of the IPTU assessment formula supports this assumption. First, the tax increase is not a market outcome. Each property's tax liability is computed from a rigid, government-set formula whose inputs are observable physical attributes: built area, land area, building age, construction standard, and location. The 2014 reform updated the formula's parameters but not its structure, so each property's tax change is a determinis-

tic function of pre-reform characteristics. Property owners had no ability to influence their assessed values or the resulting tax increase.

Second, block fixed effects and property controls absorb the key determinants of assessed values. Block fixed effects absorb land prices, which are assigned at the street level and are nearly uniform within blocks, as Figure 1 illustrates. Within ZF-1, construction prices vary only by building type, so controlling for building type absorbs the construction price component. The remaining controls, construction area, land area, and building age, absorb the other observable attributes that enter the formula. What remains after conditioning on these variables is variation generated by the nonlinear interaction of property attributes through the assessment formula. We control for construction area, land area, and building age flexibly, but the formula combines these inputs nonlinearly: two properties on the same block with similar area and age can have nearly identical assessed values yet experience different tax increases, because the reform updated land and construction price parameters by different amounts, and each property's exposure to these updates depends on the composition of its assessed value. Because this residual variation is mechanical and formula-driven, it is unlikely to reflect unobserved property quality or owner characteristics that independently predict mobility.

We provide two pieces of evidence supporting this assumption. First, we estimate the same specification using pre-reform mobility (2009–2013) as the outcome. If residual tax increases captured unobserved property characteristics that persistently predict mobility, they should predict mobility in both periods. The placebo coefficient is indistinguishable from zero in both the full sample and the RAIS subsample. Second, the main result is stable across all four control specifications (decile bins, linear, quintile bins, and B-splines), suggesting that it is not driven by a particular functional form assumption. The residualized distribution of tax increases, after absorbing block fixed effects and controls, is smooth and approximately normal, with sufficient variation for precise estimation (Appendix B.3). A binscatter of residualized tax increases against residualized mobility, constructed following Cattaneo et al. (2024), confirms that the relationship is positive and approximately linear across the distribution (Appendix Figure B.4).

4.3 Heterogeneity Specifications

A natural question is whether the mobility response varies with owner age. Lifecycle models predict that older homeowners, who face shorter horizons, declining income, and longer tenure, may respond differently to tax increases than younger owners (Coven et al., 2025). To test this, we augment the main specification with an age threshold interaction, esti-

mated on the RAIS-matched subsample of formally employed owners (the subset for whom administrative employer-employee records provide demographic information):

$$Y_i = \beta_0 \cdot \text{TaxIncrease}_i + \delta_{\bar{a}} \cdot \text{TaxIncrease}_i \times \mathbf{1}(\text{Age}_i \geq \bar{a}) + \gamma \cdot \mathbf{1}(\text{Age}_i \geq \bar{a}) + f(X_i) + \alpha_j + \varepsilon_i, \quad (4)$$

where $\bar{a}$ denotes the age threshold, β_0 captures the effect of the tax increase for owners below $\bar{a}$, and $\delta_{\bar{a}}$ measures the differential response for older owners. The threshold indicator $\mathbf{1}(\text{Age}_i \geq \bar{a})$ absorbs any discrete level shift in mobility at the cutoff, so that $\delta_{\bar{a}}$ isolates the differential tax response rather than reflecting baseline age differences in mobility. We set $\bar{a}$ at the quartiles of the owner age distribution: $\bar{a} \in \{42, 50, 58\}$, corresponding to the 25th, 50th, and 75th percentiles. All specifications on the RAIS subsample include owner controls (age, log wage, college indicator, and gender) alongside the property controls in $f(X_i)$.

5 Results

5.1 Tax Increases Raise Residential Mobility

Table 2 reports OLS estimates of equation (3) on the full sample of 295,991 ZF-1 apartments. In the main specification with decile-binned controls (column 1), apartments that experienced a 10 percentage point larger tax increase are 0.86 percentage points more likely to transact over the six years following the reform (2015–2020). The estimate is stable across all four control specifications. To put this in context, the standard deviation of the tax increase across ZF-1 apartments is 7.5 percentage points. At the estimated coefficient, apartments with a one-standard-deviation larger tax increase have a 0.65 percentage point higher six-year mobility rate, or 4.5% relative to the baseline rate of 14.4%.

Our identification assumes that residual variation in tax increases, after conditioning on block fixed effects and property controls, does not reflect pre-existing differences in residential mobility. To test this, Panel B of Table 2 replaces the dependent variable with an indicator for pre-reform transactions (2009–2013). The placebo coefficient in the main specification is small and not statistically significant, and none of the four specifications yields a significant estimate. The contrast between the precisely estimated post-reform effect and the null placebo supports a causal interpretation.

Table 3 replicates the analysis on the RAIS-matched subsample of 54,278 formally employed owners, with linear property controls as the main specification and additional controls for owner wage, education, and gender. The RAIS sample is not representative of all homeowners, as it is restricted to formal sector employees. Nevertheless, the estimated effect of tax increases on mobility remains positive and statistically significant across all four specifica-

Table 2: Effect of Tax Increase on Residential Mobility (β): Full Sample

	(1)	(2)	(3)	(4)
Panel A: Post-Reform (2015–2020)				
Tax Increase	0.086***	0.117***	0.094***	0.087***
	(0.026)	(0.025)	(0.027)	(0.026)
Dep. var. mean	0.144	0.144	0.144	0.144
Panel B: Pre-Reform Placebo (2009–2013)				
Tax Increase	-0.028	0.081	-0.024	0.013
	(0.042)	(0.054)	(0.045)	(0.049)
Dep. var. mean	0.235	0.235	0.235	0.235
Decile bins	✓			
Linear controls		✓		
Quintile bins			✓	
B-spline basis				✓
Location FE	✓	✓	✓	✓
N	295,991	295,991	295,991	295,991

Notes: Standard errors clustered at the *setor* (census tract) level in parentheses. The dependent variable in Panel A is an indicator for any ITBI market transaction during 2015–2020; in Panel B, during 2009–2013 (pre-reform placebo). All specifications include property controls (construction area, land area, land fraction, property age) in the indicated functional form and construction standard dummies. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

tions. The corresponding placebo (Panel B of Table 3) is again null across all specifications. This subsample forms the basis for the heterogeneity analysis below.

Table 3: Effect of Tax Increase on Residential Mobility (β): RAIS Subsample

	(1)	(2)	(3)	(4)
Panel A: Post-Reform (2015–2020)				
Tax Increase	0.147***	0.131**	0.130**	0.133**
	(0.053)	(0.057)	(0.056)	(0.063)
Dep. var. mean	0.138	0.138	0.138	0.138
Panel B: Pre-Reform Placebo (2009–2013)				
Tax Increase	-0.064	-0.090	-0.047	-0.090
	(0.125)	(0.102)	(0.109)	(0.119)
Dep. var. mean	0.311	0.311	0.311	0.311
Linear controls	✓			
Decile bins		✓		
Quintile bins			✓	
B-spline basis				✓
Owner controls	✓	✓	✓	✓
Location FE	✓	✓	✓	✓
N	54,278	54,278	54,278	54,278

Notes: Standard errors clustered at the *setor* (census tract) level in parentheses. The dependent variable in Panel A is an indicator for any ITBI market transaction during 2015–2020; in Panel B, during 2009–2013 (pre-reform placebo). All columns include owner controls (log wage, college indicator, female indicator). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.2 Heterogeneity by Owner Age

The RAIS subsample allows us to examine whether the mobility response varies with owner age. Table 4 reports estimates of the differential effect $\delta_{\bar{a}}$ from equation (4) at three age thresholds set at the 25th, 50th, and 75th percentiles of the owner age distribution in our sample ($\bar{a} \in \{42, 50, 58\}$). The results reveal a monotonic age gradient. At the 25th percentile ($\bar{a} = 42$) and the median ($\bar{a} = 50$), the differential effect is positive but statistically indistinguishable from zero. At the 75th percentile ($\bar{a} = 58$), the differential effect is large and stable across all four control specifications, indicating that owners aged 58 and above respond considerably more to tax increases than younger owners. Columns (5)–(8) report the same specifications using pre-reform mobility (2009–2013) as the dependent variable. The placebo estimates are uniformly small and insignificant, confirming that the differential response among older owners is specific to the post-reform period.

Table 4: Heterogeneity by Owner Age ($\delta_{\bar{a}}$)

	Post-Reform (2015–2020)				Pre-Reform Placebo (2009–2013)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\bar{a} = 42$ (P25)	0.026	0.020	0.024	0.023	-0.036	-0.055	-0.041	-0.058
	(0.064)	(0.064)	(0.065)	(0.065)	(0.072)	(0.074)	(0.074)	(0.074)
$\bar{a} = 50$ (P50)	0.049	0.041	0.046	0.043	0.046	0.009	0.026	0.007
	(0.052)	(0.052)	(0.052)	(0.053)	(0.068)	(0.068)	(0.068)	(0.068)
$\bar{a} = 58$ (P75)	0.132***	0.120***	0.123***	0.123***	0.060	0.011	0.021	0.005
	(0.045)	(0.045)	(0.045)	(0.046)	(0.069)	(0.067)	(0.068)	(0.069)
Linear controls	✓				✓			
Decile bins		✓				✓		
Quintile bins			✓				✓	
B-spline basis				✓				✓
Owner controls	✓	✓	✓	✓	✓	✓	✓	✓
Location FE	✓	✓	✓	✓	✓	✓	✓	✓
N	54,278	54,278	54,278	54,278	54,278	54,278	54,278	54,278

Notes: Each row reports $\delta_{\bar{a}}$, the differential effect of the tax increase for owners aged $\bar{a}$ and above relative to younger owners. The specification is: $Y_i = \beta_0 \cdot \text{TaxIncrease}_i + \delta_{\bar{a}} \cdot \text{TaxIncrease}_i \times \mathbf{1}(\text{Age}_i \geq \bar{a}) + \gamma \cdot \mathbf{1}(\text{Age}_i \geq \bar{a}) + f(X_i) + \alpha_j + \varepsilon_i$. Columns (1)–(4) use post-reform mobility (2015–2020) as the dependent variable; columns (5)–(8) use pre-reform mobility (2009–2013) as a placebo. Standard errors in parentheses, clustered at the *setor* (census tract) level. All columns include owner controls (age, log wage, college indicator, female indicator) in $f(X_i)$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

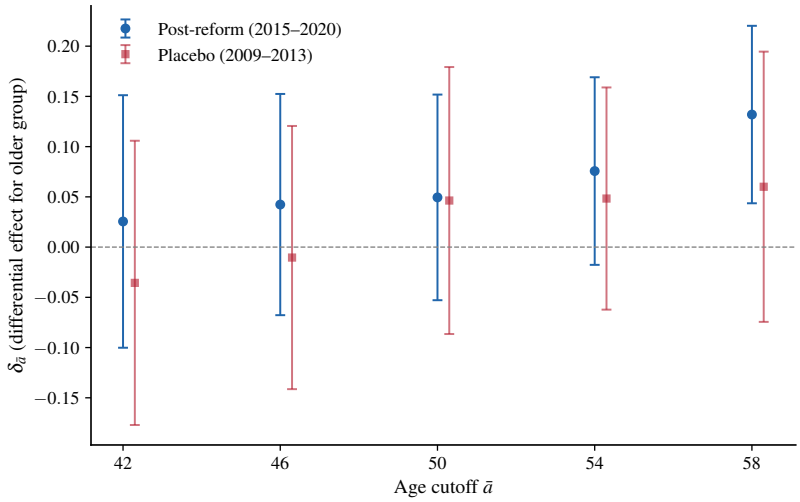


Figure 4: Older owners respond more to tax increases

Notes: Each point plots $\delta_{\bar{a}}$ from equation (4) at the indicated age threshold $\bar{a}$, with 95% confidence intervals. The specification uses linear property controls and owner controls (age, log wage, college indicator, female indicator), with standard errors clustered at the *setor* level.

The age gradient is not driven by the choice of threshold. Figure 4 traces $\delta_{\bar{a}}$ across a finer grid of age cutoffs, showing that the effect increases monotonically with age. This is consistent with Coven et al. (2025), who model how higher property taxes raise the user cost of ownership and reduce lock-in among existing homeowners. Ferreira (2010) and Wasi and White (2005) provide complementary evidence from the opposite direction: under California's Proposition 13, lower property tax burdens for long-tenure owners reduce mobility, with the lock-in effect growing as the gap between assessed and market values widens over time. Our estimates complement these findings by showing that higher property tax burdens increase mobility among older owners.

5.3 Robustness

We first examine how the mobility response evolves with the length of the post-reform window. Figure 5 re-estimates equation (3) varying the horizon over which mobility is measured. For example, a window of three years defines the dependent variable as whether the property experienced a transaction between 2015 and 2017; a pre-reform window of two years covers transactions between 2013 and 2014. The effect is small and insignificant over the first year, increases through the eight-year window. This pattern is consistent with a causal effect that accumulates as affected owners gradually move. Pre-reform windows of comparable length yield coefficients close to zero with no discernible trend, reinforcing that

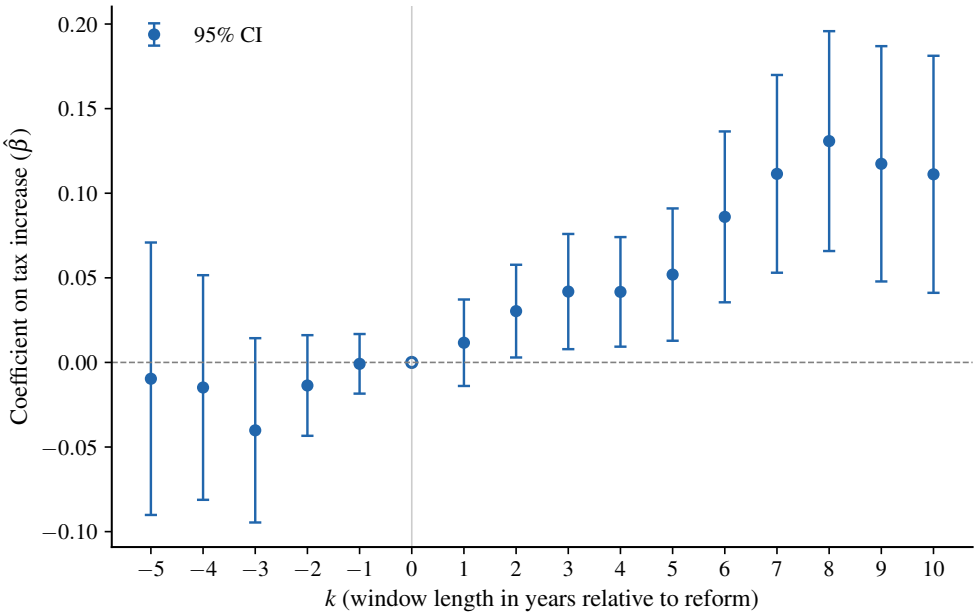


Figure 5: The effect grows with the post-reform window and stabilizes

Notes: Each point plots the OLS coefficient on the tax increase from equation (3), varying the post-reform window used to define the dependent variable. For $k > 0$, mobility is measured over $[2015, 2014 + k]$; for $k < 0$, over pre-reform windows ending at 2014. The point at $k = 0$ is normalized to zero. Error bars show 95% confidence intervals. Main specification with decile-binned controls and block fixed effects.

the estimated effect does not reflect pre-existing mobility differences.

Our main specification uses the initial 2014–2015 tax increase as the treatment variable, which is the most conservative measure of the reform's impact. São Paulo's assessment formula incorporates input prices that are updated periodically, so the gap in tax liability between more- and less-affected properties widened over time; by 2020, the cumulative tax increase is approximately 1.7 times larger than the initial change on average. Appendix B.4 replicates the main specifications using this cumulative measure. The coefficients are proportionally smaller, as expected given the larger treatment magnitude, and remain statistically significant across all four control specifications. We prefer the initial (2014–2015) measure because it isolates the reform-driven shock before subsequent administrative updates introduce additional variation.

6 Discussion

6.1 Distributional Consequences of the Reform

We now turn to the distributional consequences of the reform and interpret the mobility findings in light of the existing literature. As shown in Appendix Figure B.6a, the reform substantially increased assessed values at all market price levels, consistent with its revenue-raising objective. However, as Appendix Figure B.6b shows, effective tax rates rose proportionally more for lower-valued properties. Among owners matched to RAIS employer records, we can directly measure the relative burden of the tax as the ratio of annual tax liability to annual wage. Appendix Figure B.7 plots this ratio against log annual wage before and after the reform. The slope shifted from -0.06 to -0.10, indicating that lower-wage owners experienced a larger increase in tax burden relative to their earnings.

Using the RAIS linkage, we construct a direct measure of relative burden: the ratio of pre-reform annual tax liability to annual wage. As shown in Table 5, owners whose tax-to-wage ratio exceeds the sample median show a larger mobility response ($\hat{\delta} \approx 0.10$), with a clean placebo.

6.2 Limitations and External Validity

Our analysis has several limitations. First, we observe only the ownership market. If tax increases also affect rental markets, for example by inducing landlords to sell or by changing rents, our estimates capture only part of the housing market response. A general equilibrium analysis incorporating both rental markets is beyond the scope of this paper. Second, the RAIS-matched subsample is restricted to formal sector employees, which excludes informal workers, retirees, and the self-employed. The heterogeneity results by age and relative burden should be interpreted with this constraint in mind; future research with more comprehensive owner data would allow a fuller characterization of the mobility response. Finally, ITBI records capture formal transactions but may miss informal transfers, which would attenuate our estimates.

São Paulo offers a useful setting for studying tax-induced mobility. As the largest city in Latin America, the housing market is thick enough to provide sufficient statistical power for comparing properties at a very granular level, controlling for the provision of local public goods through block-level fixed effects. Combined with a natural experiment that generated substantial within-block variation in tax changes, this setting allows us to estimate the effect of property taxes on mobility holding local amenities constant. Our findings are therefore likely informative for large urban housing markets where property taxes represent

Table 5: Heterogeneity by Tax Burden

	Tax Burden Heterogeneity (P50 cutoff)			
	(1)	(2)	(3)	(4)
Panel A: Main (2015–2020)				
Tax increase × High burden (δ)	0.103**	0.102**	0.105**	0.108**
	(0.051)	(0.051)	(0.051)	(0.051)
Panel B: Placebo (2009–2013)				
Tax increase × High burden (δ)	-0.056	-0.065	-0.062	-0.048
	(0.067)	(0.066)	(0.065)	(0.067)
Decile bins	✓			
Linear controls		✓		
Quintile bins			✓	
B-spline basis				✓
Owner controls	✓	✓	✓	✓
Location FE	✓	✓	✓	✓
N	54,278	54,278	54,278	54,278

Notes: High burden = $\mathbf{1}$(Tax Bill/Wage $\geq$ P50), where the P50 cutoff is 0.0131. Tax bill is the pre-reform (2014) annual liability; wage is 2014 annual earnings from RAIS. δ is the differential effect for owners with above-median tax burden. Standard errors in parentheses, clustered at the *setor* (census tract) level. All columns include owner controls (age, log wage, college indicator, female indicator). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

a meaningful share of housing costs. The existing empirical evidence on tax-induced mobility comes exclusively from high-income countries: the United States (Ferreira, 2010; Wasi and White, 2005; Fraenkel, 2024; Shan, 2010; Giesecke and Mateen, 2022) and Denmark (Nielsson et al., 2024). Our results from São Paulo, the largest city in South America, demonstrate that tax-induced mobility operates in a middle-income country context as well, extending the external validity of the existing evidence base.

6.3 Conclusion

This paper estimates the causal effect of property tax increases on residential mobility using São Paulo's 2014 reform. São Paulo's thick housing market, combined with a reform that generated differential tax changes among properties on the same city block, allows us to estimate this effect without the endogeneity that arises when tax variation reflects differences in the provision of local public goods. We find that property tax increases induce residential mobility: a 10 percentage point tax increase raises the six-year transaction probability by 0.86 percentage points, or 4.5 percent relative to the baseline rate. Linking property records to matched employer-employee data, we show that the effect is heterogeneous across owner characteristics. Older owners are the most responsive, a pattern consistent with the lifecycle and lock-in mechanisms studied in the existing literature. Our results provide the first causal evidence on which owners respond to property tax increases and the first evidence on tax-induced mobility in a developing country.

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A Assessment Formula and Tax Rate Schedules

The main text presents a simplified representation of the assessment formula. The full computation includes additional adjustment factors for lot depth, terrain characteristics (corner lots, interior lots, irregular shapes), building obsolescence, condominium ownership structure, and penalties for underutilized land. These factors interact nonlinearly to produce the final appraised value (*valor venal do imóvel*). For complete documentation of the formula, see Scot et al. (2023) or the city government's website.[6]

The key point for our identification strategy is that all adjustment factors are determined by a rigid, government-set formula based on observable property characteristics. Property owners cannot manipulate any component of their assessed values.

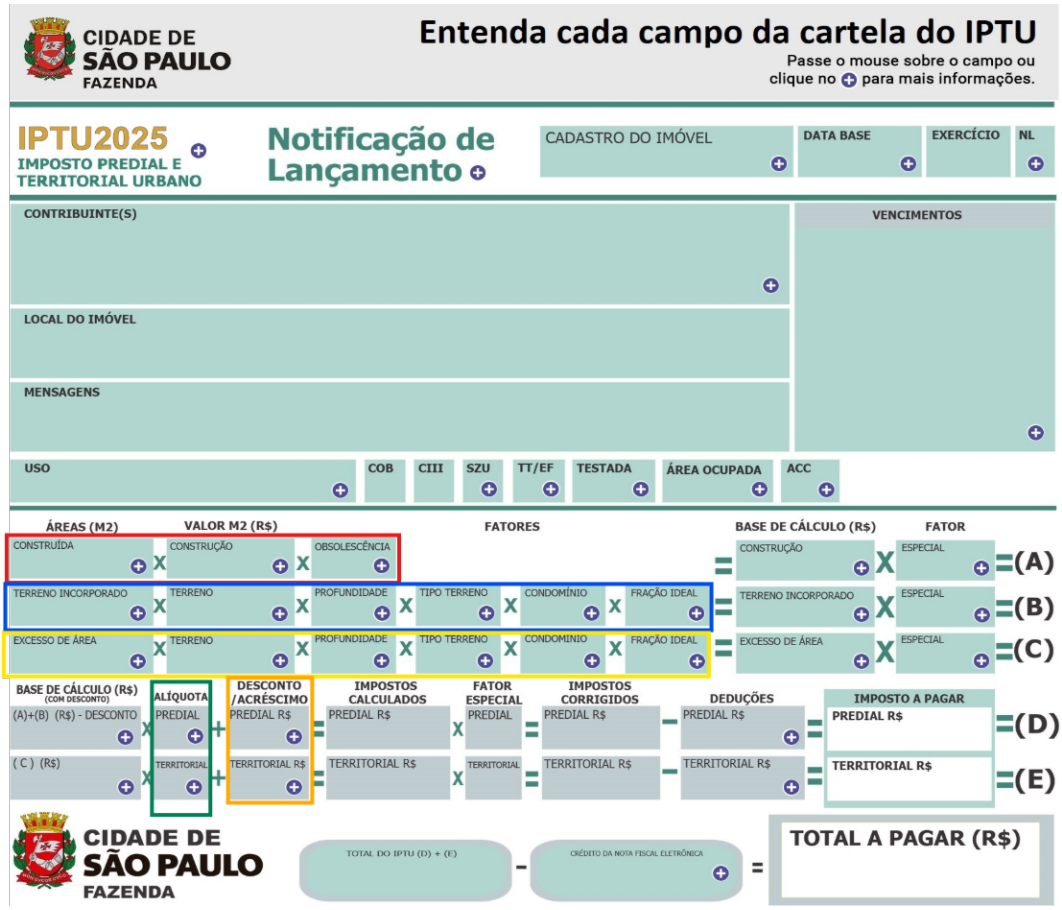


Figure A1: IPTU tax form (*Cartela do IPTU*)

Notes: Pre-filled tax form issued by the city of São Paulo. Row (A) computes the construction component: built area × construction price/m^2 × obsolescence factor. Row (B) computes the land component: lot area × land price/m^2 × adjustment factors (depth, terrain type, condominium, ideal fraction). Row (D) applies the progressive tax rate to the total assessed value. Source: Prefeitura de São Paulo.

[6] See https://web1.sf.prefeitura.sp.gov.br/CartelaIPTU/.

Once the appraised value V_i is determined, the tax liability is computed by applying a progressive rate schedule. The pre-reform (2014) schedule for residential properties was:

Appraised Value (BRL)	Marginal Rate	Fixed Deduction (BRL)
$\leq$ 77,500	0.8%	0
77,501 – 155,000	1.0%	155
155,001 – 310,000	1.2%	465
310,001 – 620,000	1.4%	1,085
$>$ 620,000	1.6%	2,325

Table A1: Pre-reform residential tax rate schedule (2014)

The post-reform (2015) schedule adjusted both the brackets and rates:

Appraised Value (BRL)	Marginal Rate	Fixed Deduction (BRL)
$\leq$ 150,000	0.7%	0
150,001 – 300,000	0.9%	300
300,001 – 600,000	1.1%	900
600,001 – 1,200,000	1.3%	2,100
$>$ 1,200,000	1.5%	4,500

Table A2: Post-reform residential tax rate schedule (2015)

The tax liability is computed as:

$$\tau_i = r_j \cdot V_i - D_j, \tag{5}$$

where r_j is the marginal rate and D_j is the fixed deduction for the bracket j containing V_i. The post-reform schedule roughly doubled the bracket thresholds while slightly lowering the marginal rates, resulting in lower effective rates for a given assessed value. However, because the reform simultaneously increased assessed values substantially, most properties experienced net tax increases.

B Additional Figures and Tables

B.1 Within-Block Variation in Land Prices

Figure B.1 shows the distribution of within-block variation in land prices across ZF-1 blocks. In 44 percent of blocks, all apartments share the same land price (SD/mean = 0). The median within-block SD/mean is 0.5 percent, and 68 percent of blocks have within-block variation

below 2 percent. This confirms that land prices are nearly uniform within blocks, so within-block variation in tax increases is driven by differences in other property characteristics interacting with the nonlinear formula, not by land price differences.

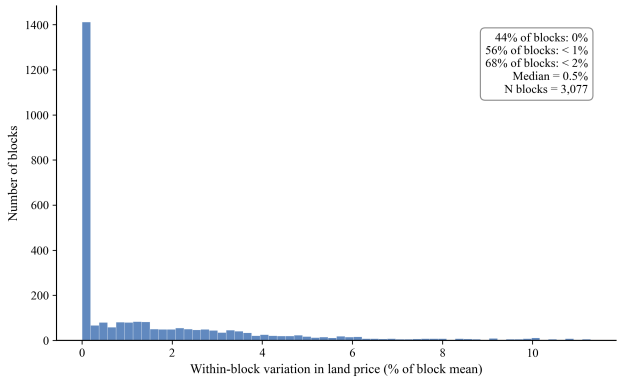
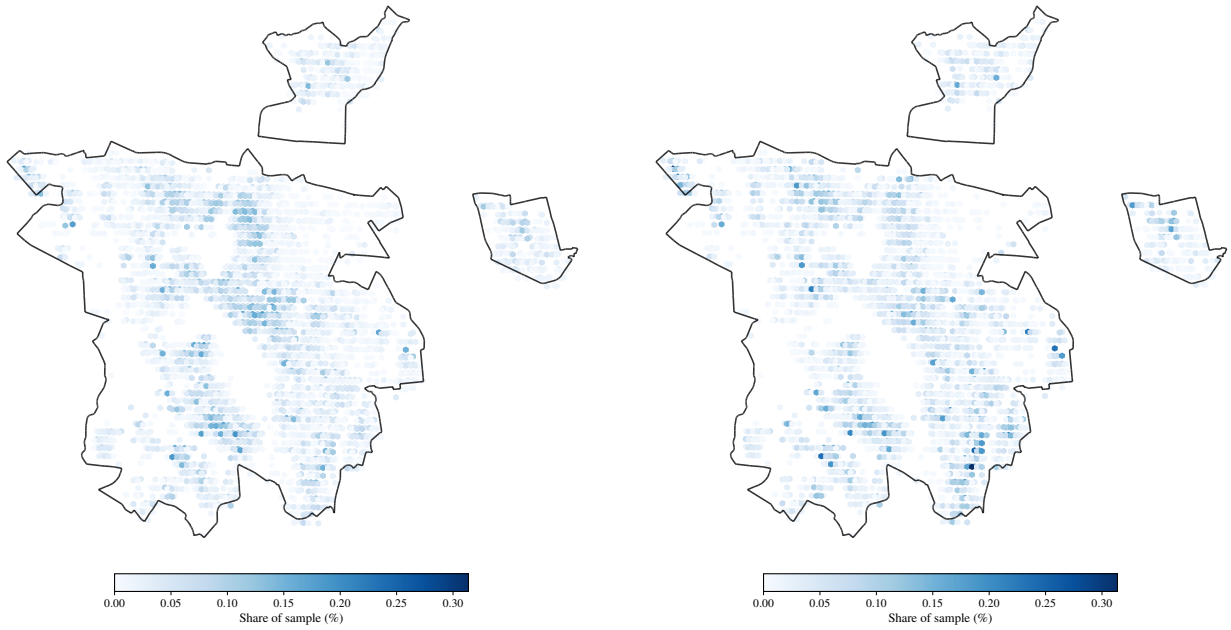


Figure B.1: Within-block variation in land prices is minimal

Notes: Distribution of within-block SD/mean of land prices ($\phi^L_{b(i)}$) across 3,077 blocks in ZF-1. 44 percent of blocks have zero variation; median SD/mean is 0.5 percent.

B.2 Sample Validation

Figure B.2 compares the spatial distribution of all ZF-1 apartments with the RAIS-matched subsample. The two distributions are nearly identical, indicating that the matched subsample is not concentrated in particular neighborhoods.



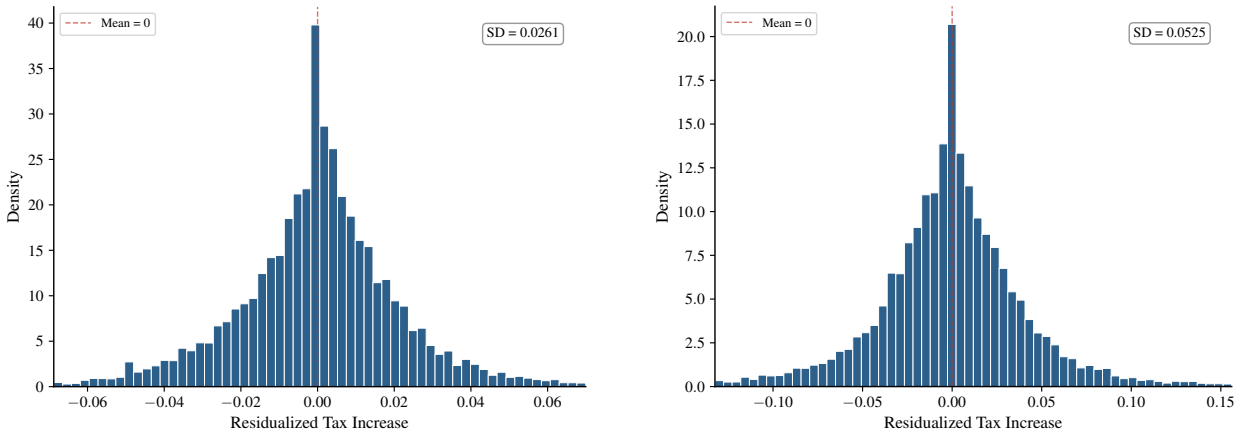
(a) All apartments ($N = 296{,}000$) (b) RAIS-matched apartments ($N = 54{,}000$)

Figure B.2: The RAIS subsample is spatially representative of the full sample

Notes: Each hexbin is colored by the number of apartments it contains. The two panels exhibit nearly identical spatial patterns, indicating that the RAIS subsample is geographically representative of the full sample.

B.3 Residualized Tax Increase Distribution

Figure B.3 shows the distribution of residualized tax increases after absorbing block fixed effects and property controls. Panel (a) displays the initial (2014–2015) measure and panel (b) displays the cumulative (2014–2020) measure. Both distributions are smooth and approximately normal, consistent with identifying variation driven by many small differences in property attributes.



(a) Initial tax increase (2014–2015)

(b) Cumulative tax increase (2014–2020)

Figure B.3: Residual tax increase variation remains after absorbing block effects

Notes: Distribution of tax increases after absorbing block (setor-by-quadra) fixed effects and property controls (decile bins of construction area, land area, property age, land fraction, and construction standard). P1–P99 trimmed.

A raw binscatter of tax increases against mobility would show the unconditional relationship, conflating the effect of the tax increase with correlations between tax increases and observable property and location characteristics (Cattaneo et al., 2024). Figure B.4 addresses this by presenting a covariate-adjusted binscatter, partialing out block fixed effects and property controls so that the figure isolates the same conditional variation that identifies the regression coefficient in Table 2. The relationship is positive and approximately linear across the distribution, with a slope consistent with the OLS estimate.

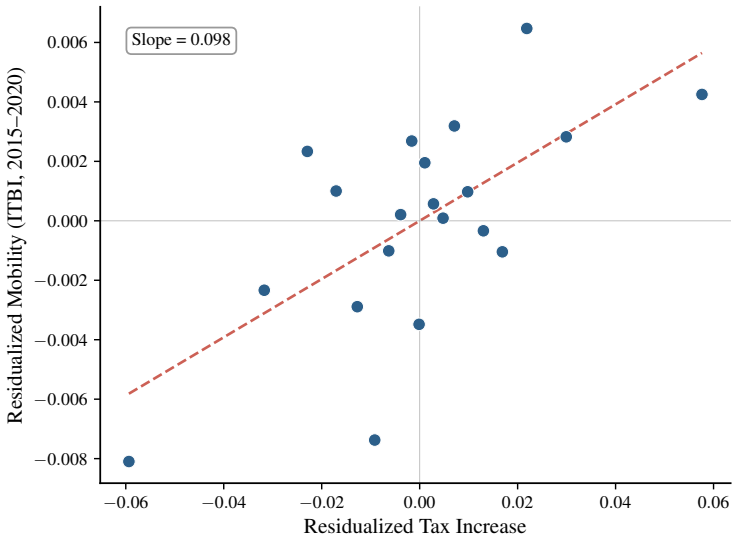


Figure B.4: Tax increases and mobility are positively correlated after absorbing block effects

Notes: Binscatter of the tax increase (x-axis) against post-reform mobility (y-axis), after partialing out decile-binned property controls and block fixed effects, constructed using the `binsreg` package (Cattaneo et al., 2024). Each point represents the conditional mean within one of 20 quantile-spaced bins. The fitted line corresponds to the OLS coefficient in the main specification of Table 2.

B.4 Cumulative Tax Increase (2014–2020)

The main specifications use the initial tax increase from 2014 to 2015 as the treatment variable. As a robustness check, we replicate the full-sample results using the cumulative tax increase from 2014 to 2020, defined as $(\tau_{2020} - \tau_{2014})/\tau_{2014}$. The cumulative measure reflects continued adjustment of assessment parameters after the initial reform, as well as changes in building obsolescence factors between 2015 and 2020.

Table B.1 reports the results. The coefficients are smaller than those in the main specification, consistent with the cumulative measure being approximately 1.7 times larger than the initial measure on average, as Figure B.5 shows. We prefer the initial (2014–2015) measure because it cleanly isolates the reform-driven shock. The cumulative results nonetheless confirm that the mobility response is not an artifact of the particular treatment definition.

Table B.1: Mobility and cumulative tax increase (2014–2020)

	(1)	(2)	(3)	(4)
Tax Increase	0.038***	0.061***	0.050***	0.037***
	(0.012)	(0.011)	(0.011)	(0.013)
Decile bins	✓			
Linear controls		✓		
Quintile bins			✓	
B-spline basis				✓
Location FE	✓	✓	✓	✓
Dep. var. mean	0.144	0.144	0.144	0.144
N	295,991	295,991	295,991	295,991

Notes: Standard errors clustered at the *setor* (census tract) level in parentheses. The dependent variable is an indicator for any ITBI market transaction during 2015–2020. The treatment variable is the cumulative tax increase from 2014 to 2020: $(\tau_{2020} - \tau_{2014})/\tau_{2014}$. All specifications include property controls (construction area, land area, land fraction, property age) in the indicated functional form and construction standard dummies. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

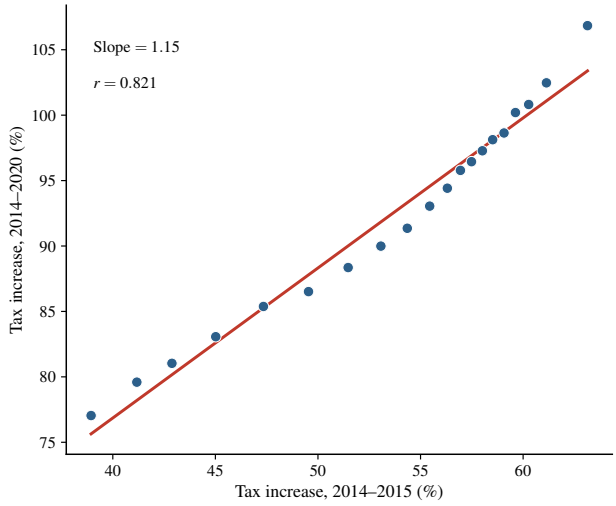
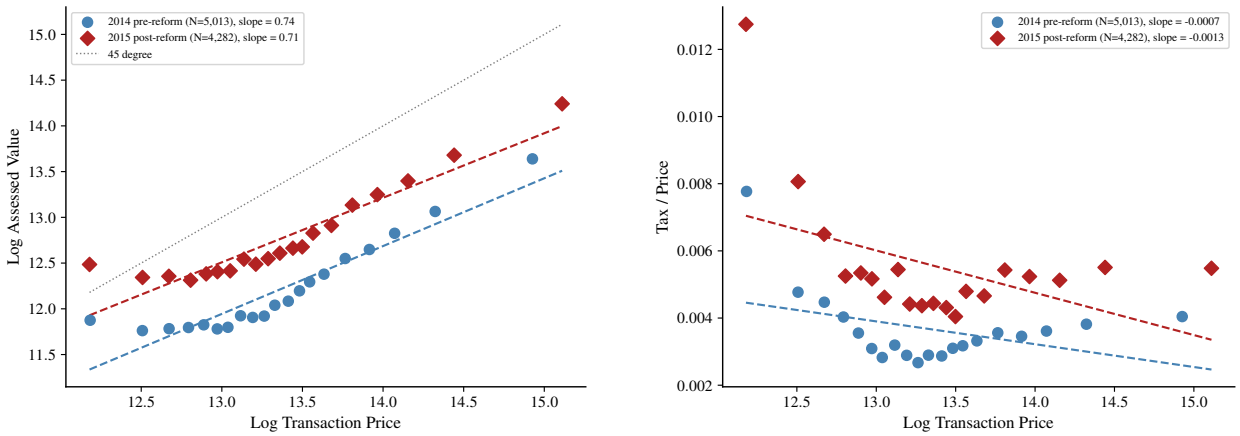


Figure B.5: Cumulative tax increases grow over time as inflation adjustments compound

Notes: Each dot is a vigintile (5% bin) mean. The slope of 1.15 indicates that cross-sectional tax differences created by the reform grew over time. The cumulative (2014–2020) increase is approximately 1.7 times larger than the initial (2014–2015) increase on average.

B.5 Distributional Consequences of the Reform



(a) Assessed value vs. transaction price

(b) Effective tax rate vs. transaction price

Figure B.6: The reform increased effective tax rates more for lower-valued properties

Notes: Panel (a) plots log assessed value against log transaction price for ZF-1 apartments transacted in 2014 (pre-reform) and 2015 (post-reform). Panel (b) plots the effective tax rate (annual tax liability / transaction price) against log transaction price. Each point represents the unconditional mean within a bin of log transaction price.

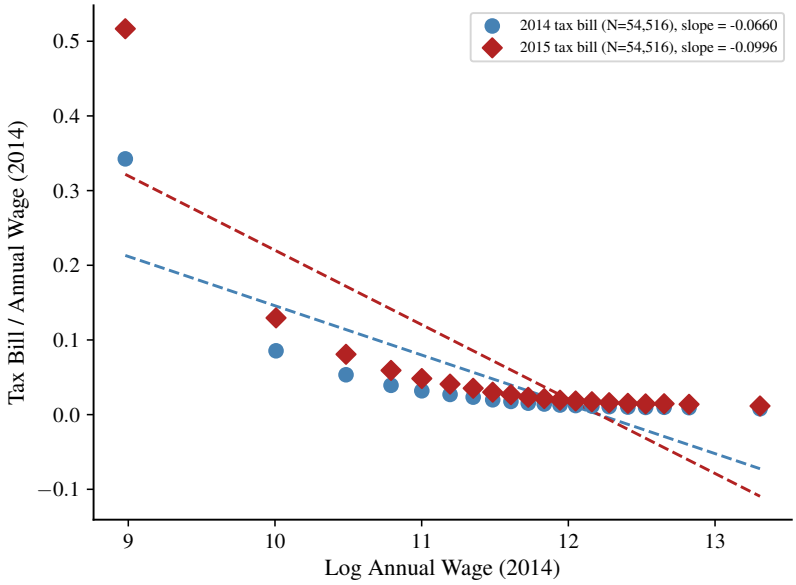


Figure B.7: The reform increased the tax-to-wage ratio more for lower-wage owners

Notes: Each point plots the mean ratio of annual IPTU tax liability to annual wage (from RAIS) within a bin of log annual wage, separately for pre-reform (2014) and post-reform (2015) tax schedules. The lines are linear fits. The sample is restricted to property owners matched to RAIS employer records ($N \approx 54{,}000$).